

Gosford High School

2014

TRIAL
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

- **General Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 6 – 15

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10.

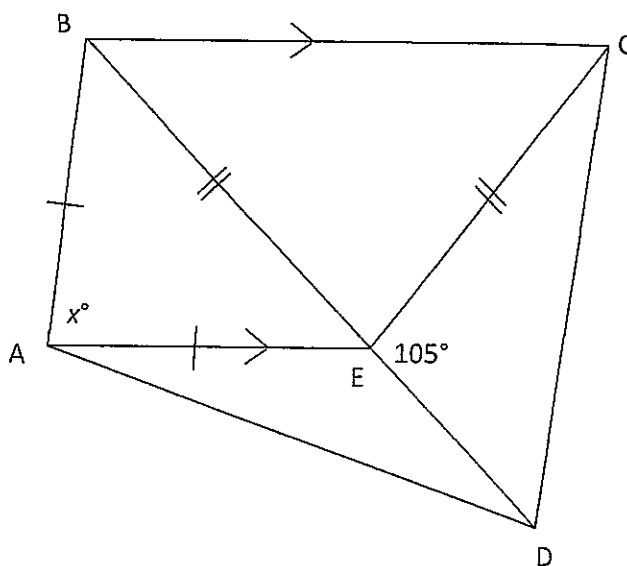
Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. An engine under development contains three components that are prone to failure. The probabilities that each of the components will fail are $\frac{1}{10}$, $\frac{1}{5}$ and $\frac{1}{4}$ respectively. What is the probability that at least one part will fail?

- (A) $\frac{23}{50}$
- (B) $\frac{1}{200}$
- (C) $\frac{3}{100}$
- (D) $\frac{11}{20}$

2. The vertices of quadrilateral ABCD meet at E such that $BC \parallel AE$, BE is produced to D. $\angle CED = 105^\circ$, $BE = CE$ and $AB = AE$. Determine the size of x .



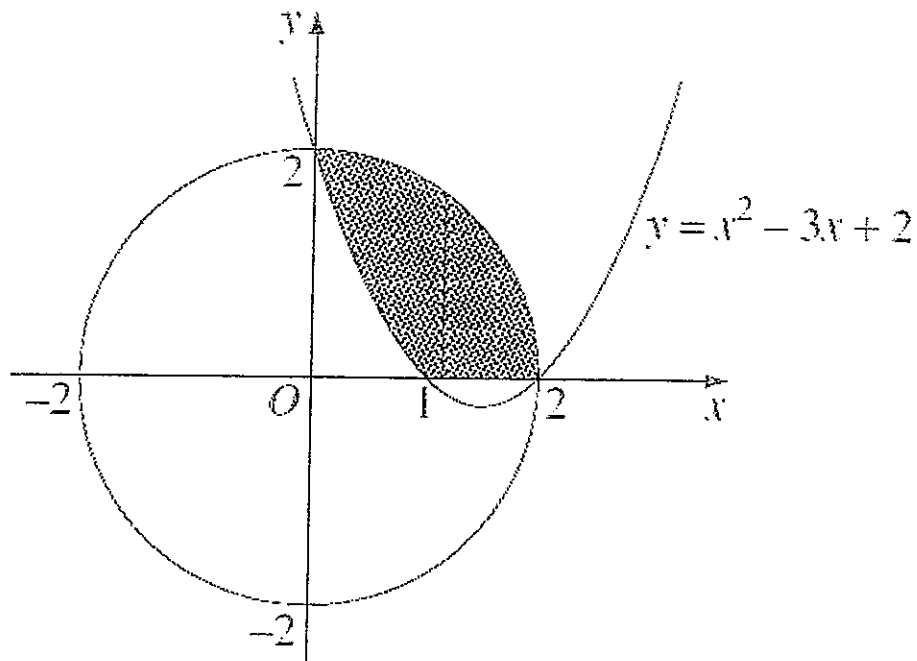
- (A) 105°
- (B) 85°
- (C) 75°
- (D) 52.5°

3. Fully simplify the algebraic fraction: $\frac{x^3 - 8}{x^2 - 4}$.
- (A) $\frac{x^2 - 2x + 4}{x - 2}$
- (B) $x + 2$
- (C) $\frac{x^2 + 4x + 4}{x + 2}$
- (D) $\frac{x^2 + 2x + 4}{x + 2}$
4. What is the value of $\frac{dy}{dx}$ if $y = 4\sqrt{x}$?
- (A) $\frac{dy}{dx} = 4$
- (B) $\frac{dy}{dx} = 2\sqrt{x}$
- (C) $\frac{dy}{dx} = \frac{2}{\sqrt{x}}$
- (D) $\frac{dy}{dx} = -\frac{1}{2\sqrt{x}}$
5. During a two year study, the number of people using public transport increased by 6% in the first year and in the following year the number increased by a further 14% to 680 000. The number of people using public transport at the beginning of the study was approximately:
- (A) 8 095
- (B) 549 712
- (C) 562 728
- (D) 544 000
6. The quadratic function $3x^2 - 6x - 7$ has roots α and β .
What is the value of $\alpha^2 + \beta^2$?
- (A) $\frac{26}{3}$
- (B) $-\frac{7}{3}$
- (C) 4
- (D) 45

7. The primitive function of $\sec^2 5x + 1$ is:

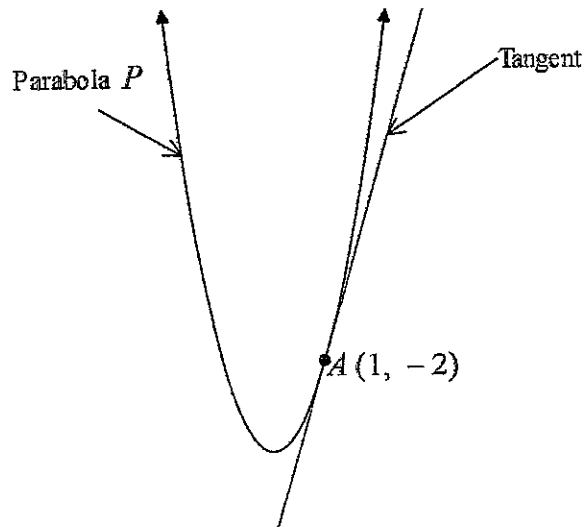
- (A) $\frac{1}{5}\tan(5x) + x + C$
- (B) $\frac{1}{5}\tan(x) + 5x + C$
- (C) $\tan(5x) + x + C$
- (D) $\tan(5x) + C$

8. Which set of inequalities best describes the shaded region.



- (A) $x^2 + y^2 \leq 2$ and $y \geq x^2 - 3x + 2$ and $y \geq 0$
- (B) $x^2 + y^2 \leq 4$ and $y \leq x^2 - 3x + 2$ and $x \geq 0$
- (C) $x^2 + y^2 \leq 2$ and $y \leq x^2 - 3x + 2$ and $y \geq 0$
- (D) $x^2 + y^2 \leq 4$ and $y \geq x^2 - 3x + 2$ and $x \geq 0$

9. The diagram shows the parabola P and its tangent at the point $A(1, -2)$.



Which of the following equations might represent the normal to the parabola at the point A ?

- (A) $x - 3y + 5 = 0$
(B) $2x - 3y + 1 = 0$
(C) $x + 3y + 5 = 0$
(D) $x + 3y - 5 = 0$
10. Differentiate $\log_e \left(\frac{x-5}{x+5} \right)$

- (A) $\frac{25}{x^2-25}$
(B) $\frac{10}{x^2-25}$
(C) $\frac{1}{x-5} + \frac{1}{x+5}$
(D) $\frac{1}{x+5} - \frac{1}{x-5}$

Section II

90 marks

Attempt Questions 11 – 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start on a new page.

- (a) Evaluate $\log_{10} 12$ correct to 3 significant figures. 1
- (b) Solve $|2x - 3| = 4 - 3x$ 2
- (c) Differentiate the following functions with respect to x :
- (i) $3xe^x$ 1
- (ii) $\frac{1}{(x^2+5)^3}$ 1
- (iii) $\frac{x^2+4}{3x-1}$ 1
- (d) If $f'(x) = 6x^2 + 5x - 1$ and $f(-1) = 5$, find an expression for $f(x)$. 2
- (e) A particle moves so that its displacement from the origin is given by:
- $$x = -t^2 + 7t + 8 \quad (\text{where } x \text{ is displacement in metres and } t \text{ is time in seconds})$$
- (i) What is the initial displacement of the particle? 1
- (ii) At what time will the particle be at the origin? 2
- (f) Consider the arithmetic sequence beginning with $-7, -3, 1 \dots$:
- (i) Find the 25th term in the sequence. 1
- (ii) Show that the sum of the first 20 terms is 620 1
- (iii) How many terms must be taken to give a sum of 221? 2

End of Question 11

Question 12 (15 marks) Start on a new page.

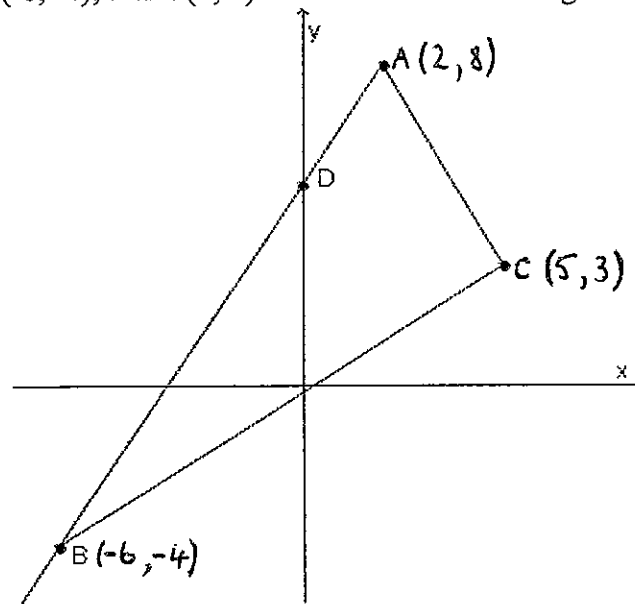
- (a) Sammy sailed 15km south from point A to point B. He then sailed due west to C where he was on a bearing of 210° from A.

(i) Draw a diagram showing this information. 1

(ii) Determine the exact distance AC 2

(iii) Sammy then sailed NW for 8km to D. What is the distance from A to D correct to the nearest 100m. 2

- (b) The points $A(2, 8)$, $B(-6, -4)$, and $C(5, 3)$ are the vertices of triangle ABC.



(i) Find the equation of the line passing through A and B. 2

(ii) Find the coordinates of the point D. 1

(iii) Find the perpendicular distance from point C to the line AB. 2

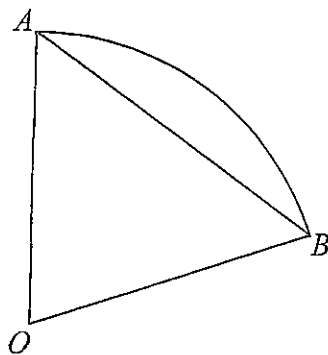
(iv) Find the exact distance between A and B. 2

(v) Calculate the area of $\triangle ABC$. 1

Question 12 continues on page 9

Question 12 (continued)

- (c) Sector AOB has arc length $AB = 2\pi$ units and radii $OA = OB = 6$ units.



- (i) Determine the size, in radians, of angle $A\hat{O}B$. 1
- (ii) Show that the area of $\triangle AOB$ is $9\sqrt{3}$ square units. 1

End of Question 12

Question 13 (15 marks) Start on a new page.

(a) A function is given by $f(x) = -x^3 + \frac{3x^2}{2} + 6x$

(i) Find the coordinates of the stationary points of $f(x)$ and determine their nature. 3

(ii) Find the coordinates of the point of inflexion of $f(x)$. 1

(iii) Hence, sketch the graph $y = f(x)$ showing the stationary points, point of inflexion and y intercept. 3

(iv) For what values of x is the function decreasing? 1

(b) Consider the function $f(x) = 4 \sin 2\pi x$

(i) Determine the period and amplitude of $f(x)$ 2

(ii) Determine the values of x for which $f(x)$ is at its maximum in the domain $0 \leq x \leq 2$ 2

(iii) Sketch $f(x)$ for $0 \leq x \leq 2$. 1

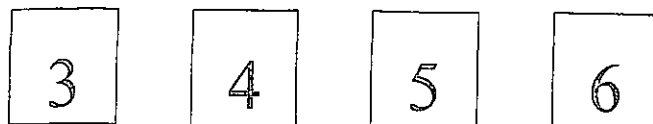
(c) Evaluate: $\lim_{x \rightarrow 2} \frac{3x^2 - 12}{x - 2}$ 2

End of Question 13

Question 14 (15 marks) Start on a new page.

- (a) Use Simpson's rule to approximate $\int_1^5 \frac{dx}{x^2+1}$, using 4 sub-intervals to 4 decimal places. 3

- (b) Four cards, numbered 3, 4, 5 and 6 are used in a game.



The four cards are placed face down and each player pays \$1 to take a turn to draw two cards, one at a time without replacement. The two cards make a 2-digit number, with the first card drawn being the first digit of their number.

If the cards form a number over 60, the player receives \$3 back (i.e. they win \$2), otherwise they receive nothing back.

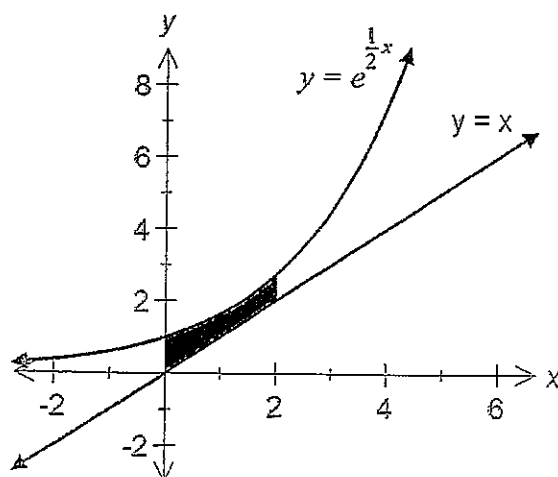
The four cards are then shuffled and replaced for the next turn.

- (i) Use a diagram to show all of the possible 2-digit numbers that could be drawn. 2

- (ii) What is the probability that a player will win on their first turn? 1

- (iii) Calculate the probability that a player who brings \$5 to play will have \$3 left after 5 turns? 2

- (c) The diagram shows the graphs of the functions $y = e^{\frac{1}{2}x}$ and $y = x$. The region between these 2 functions and the bounds $x = 0$ and $x = 2$ has been shaded. 3



Calculate the exact area of the shaded region.

Question 14 continues on page 12

Question 14 (continued)

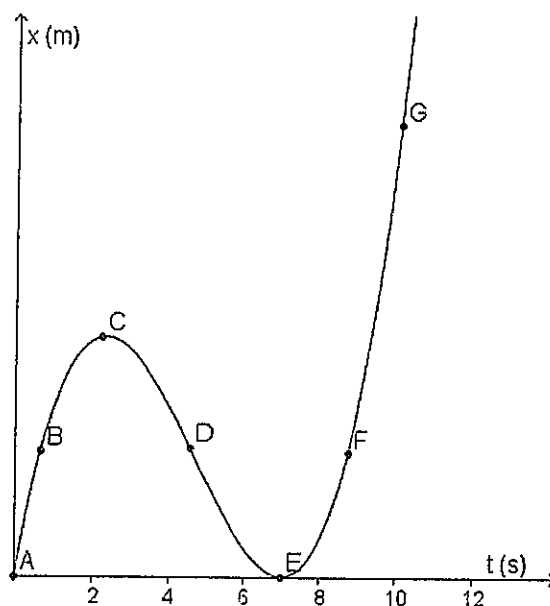
(d) For the parabola with equation $16y = x^2 - 4x - 12$:

- | | | |
|-------|-------------------------------------|---|
| (i) | Find the coordinates of the vertex. | 2 |
| (ii) | Find the coordinates of the focus. | 1 |
| (iii) | Find the equation of the directrix. | 1 |

End of Question 14

Question 15 (15 marks) Start on a new page.

- (a) The population of New South Wales in 2009 was 7.13 million. In 2012 the population had grown to approximately 7.3 million people.
- (i) Assuming that the growth rate is proportional to the population, show that the annual growth rate is approximately 0.79%. 2
- (ii) Calculate the expected population of New South Wales in 2019 using this model. Give your answer rounded to the nearest hundred thousand. 1
- (iii) In what year will the population exceed 10 million? 2
- (b) The graph shows the displacement (x metres) of a particle at time (t seconds). The particle is moving horizontally.



- (i) At which point, during the first 8 seconds is the particle at its maximum distance from the origin? 1
- (ii) Describe the motion of the particle at 4 seconds in terms of its displacement and velocity. 2
- (iii) By referring to the points marked on the graph, between what times is the acceleration of the particle negative? What feature of the graph tells us this? 2

Question 15 continues on page 14

Question 15 (continued)

- (c) Find the solutions to the equation:

3

$$4 \cos^2 \theta = 6 \sin \theta + 6 \text{ in the domain } 0 \leq \theta \leq 2\pi$$

- (d) Show that $\frac{1}{\sqrt{n} + \sqrt{n+1}} = \sqrt{n+1} - \sqrt{n}$ for all integers $n \geq 1$.

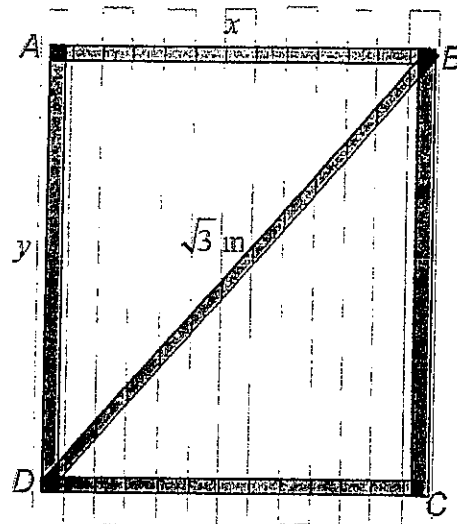
2

End of Question 15

Question 16 (15 marks) Start on a new page.

- (a) A swinging gate is to be constructed from timber palings. It will require a support frame using 5 pieces of timber: AB , AD , BD , BC and CD .

$AB \parallel CD$ and $AD \parallel BC$. $AB = CD = x$ metres. $AD = BC = y$ metres. BD is $\sqrt{3}$ metres long.



- (i) Find an expression for y in terms of x .

1

- (ii) Show that the total length (L) of the timber pieces in the support frame is

represented by $L = 2\left(x + \sqrt{3 - x^2} + \frac{\sqrt{3}}{2}\right)$.

1

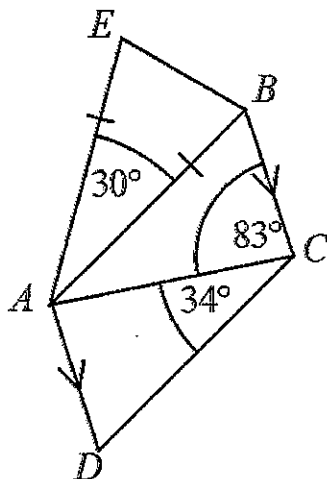
- (iii) The gate will have its maximum strength when the total length (L) of its support frame is maximised. For what value of x will the gate have maximum strength?
Full working must be shown.

4

Question 16 continues on page 16

Question 16 (continued)

- (b) In the diagram below: $AD \parallel BC$, $AE = AB$, $\angle BAE = 30^\circ$, $\angle BCA = 83^\circ$, $\angle ACD = 34^\circ$, $\angle EBC = 138^\circ$.



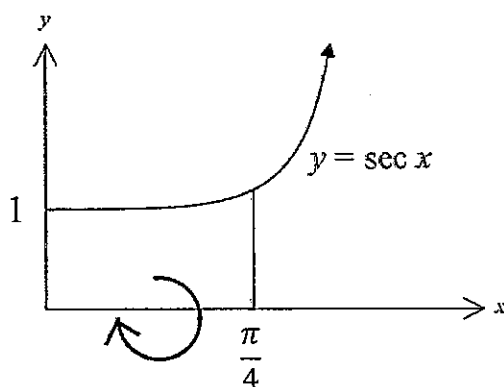
- (i) Prove that $AB \parallel DC$.

3

- (ii) Prove that $\triangle ABC \equiv \triangle ACD$.

3

- (c) The area bounded by the function $y = \sec x$, the y -axis and the line $x = \frac{\pi}{4}$ is rotated about the x -axis.



Find the volume of the solid formed.

3

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Trial HSC Examination 2014

Mathematics Course

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

 A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

 A ☒ B ☒ ^{correct} C ☐ D ☐

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

$$\begin{aligned}
 1. \quad P(\geq 1 \text{ fail}) &= 1 - P(0 \text{ fail}) \\
 &= 1 - \left(\frac{9}{10} \times \frac{4}{5} \times \frac{3}{4} \right) \\
 &= 1 - \frac{27}{50} \\
 &= \frac{23}{50} \quad (A)
 \end{aligned}$$

$$\begin{aligned}
 2. \quad \angle EBC &= \angle ECB \quad (\text{Equal base angles of isosceles } \triangle EBC \text{ opposite equal sides}) \\
 \therefore 2\angle EBC &= 105 \quad (\text{Exterior angle of } \triangle EBC) \\
 \angle EBC &= 52.5^\circ
 \end{aligned}$$

$$\begin{aligned}
 \angle AEB &= \angle CBE \quad (\text{Alternate angles on } BC \parallel AE \text{ are equal}) \\
 &= 52.5^\circ
 \end{aligned}$$

$$\begin{aligned}
 \angle ABE &= \angle AEB \quad (\text{Equal base angles of isosceles } \triangle ABE \text{ opposite equal sides}) \\
 &= 52.5^\circ
 \end{aligned}$$

$$\begin{aligned}
 x + 52.5 + 52.5 &= 180 \quad (\text{Angle sum of } \triangle) \\
 x &= 75^\circ
 \end{aligned}$$

(C)

$$\begin{aligned}
 3. \quad \frac{x^3 - 8}{x^2 - 4} &= \frac{(x-2)(x^2 + 2x + 4)}{(x+2)(x-2)}
 \end{aligned}$$

$$= \frac{x^2 + 2x + 4}{x+2} \quad (D)$$

$$\begin{aligned}
 4. \quad y &= 4x^{\frac{1}{2}} \\
 \frac{dy}{dx} &= \frac{1}{2} \times 4x^{-\frac{1}{2}} \\
 &= \frac{2}{\sqrt{x}} \quad (C)
 \end{aligned}$$

$$5. \quad \text{Number} \times 1.06 \times 1.14 = 680\,000$$

$$\text{number} = \frac{680\,000}{1.06 \times 1.14}$$

$$= 562\,727.57\dots$$

$$\approx 562\,728 \quad (C)$$

$$6. \quad 3x^2 - 6x - 7$$

$$\begin{aligned}
 x + \beta &= -\frac{b}{a} & x\beta &= \frac{c}{a} \\
 &= -\frac{-6}{3} & &= -\frac{7}{3}
 \end{aligned}$$

$$= 2$$

$$\begin{aligned}
 x^2 + \beta^2 &= (x + \beta)^2 - 2x\beta \\
 &= (2)^2 - 2\left(-\frac{7}{3}\right)
 \end{aligned}$$

$$= 4 + \frac{14}{3}$$

$$= \frac{26}{3} \quad (A)$$

$$7. \quad \int \sec^2 5x + 1 \, dx$$

$$= \frac{1}{5} \tan 5x + x + c \quad (A)$$

$$8. \quad (D)$$

$$9. \quad (C)$$

$$10. \quad \frac{d}{dx} \ln\left(\frac{x-5}{x+5}\right) = \frac{d}{dx} \ln(x-5) - \ln(x+5)$$

$$= \frac{1}{x-5} - \frac{1}{x+5}$$

$$= \frac{x+5 - x+5}{(x-5)(x+5)} = \frac{10}{x^2 - 25} \quad (B)$$

11

$$a) \log_{10} 12 = 1.07918... \\ = 1.08 \text{ (3 sig. figs.)}$$

$$b) |2x-3| = 4-3x$$

$$2x-3 = 4-3x \quad \text{or} \quad -(2x-3) = 4-3x \\ 5x = 7 \quad \quad \quad -2x+3 = 4-3x \\ x = \frac{7}{5} \quad \quad \quad x = 1$$

$$\text{Check} \quad \text{LHS} = |2(\frac{7}{5})-3| \quad \quad \text{LHS} = |2(1)-3| \\ = \frac{1}{5} \quad \quad \quad = 1 \\ \text{RHS} = 4-3(\frac{7}{5}) \quad \quad \text{RHS} = 4-3(1) \\ = -\frac{1}{5} \quad \quad \quad = 1 \\ \text{LHS} \neq \text{RHS} \quad \quad \quad \text{LHS} = \text{RHS.}$$

$\therefore x=1$ is the only solution.

$$c) (i) \frac{d}{dx} (3xe^x) = 3xe^x + 3e^x \\ = 3e^x(x+1)$$

$$(ii) \frac{d}{dx} \left(\frac{1}{(x^2+5)^3} \right) = \frac{d}{dx} (x^2+5)^{-3} \\ = -3(x^2+5)^{-4} \times 2x \\ = \frac{-6x}{(x^2+5)^4}$$

$$(iii) \frac{d}{dx} \left(\frac{x^2+4}{3x-1} \right) = \frac{(3x-1) \times 2x - 3(x^2+4)}{(3x-1)^2} \\ = \frac{6x^2 - 2x - 3x^2 - 12}{(3x-1)^2}$$

$$= \frac{3x^2 - 2x - 12}{(3x-1)^2}$$

$$d) f'(x) = 6x^2 + 5x - 1 \\ f(x) = \frac{6x^3}{3} + \frac{5x^2}{2} - x + C \\ = 2x^3 + \frac{5}{2}x^2 - x + C$$

$$f(-1) = 5 \\ 5 = 2(-1)^3 + \frac{5}{2}(-1)^2 - (-1) + C$$

$$5 = -2 + \frac{5}{2} + 1 + C \\ 5 = 1\frac{1}{2} + C \\ C = 3\frac{1}{2}$$

$$\therefore f(x) = 2x^3 + \frac{5}{2}x^2 - x + \frac{7}{2}$$

e) (i) Initial displacement $t=0$

$$x = -(0)^2 + 7(0) + 8 \\ x = 8$$

(ii) Particle at origin $\Rightarrow x=0$

$$0 = -t^2 + 7t + 8 \\ 0 = t^2 - 7t - 8 \\ 0 = (t-8)(t+1) \\ t = -1, 8$$

But $t \geq 0 \quad \therefore t = 8 \text{ seconds}$

f) $a = -7$

$d = 4$

(i) $T_n = a + (n-1)d$

$T_{25} = -7 + (25-1) \times 4$
 $= 89$

(ii) $S_n = \frac{n}{2} [2a + (n-1)d]$

$S_{20} = \frac{20}{2} [2(-7) + (20-1)(4)]$
 $= 620$

(iii) $S_n = \frac{n}{2} [2a + (n-1)d]$

$221 = \frac{n}{2} [2(-7) + (n-1)(4)]$

$442 = n(-14 + 4n - 4)$

$442 = 4n^2 - 18n$

$4n^2 - 18n - 442 = 0$

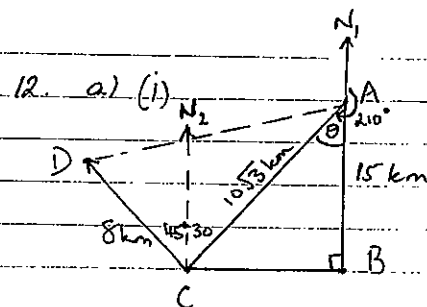
$2n^2 - 9n - 221 = 0$

$(n-13)(2n+17) = 0$

$n = 13, -\frac{17}{2}$

but $n > 0$

$\therefore n = 13$



(ii) Let $\theta = \angle CAB$

$\theta + 180 = 210^\circ$

$\theta = 30^\circ$

$\cos 30 = \frac{15}{AC}$

$\frac{\sqrt{3}}{2} = \frac{15}{AC}$

$AC = \frac{2 \times 15}{\sqrt{3}}$

$= \frac{30}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$

$= \frac{30\sqrt{3}}{3}$

$= 10\sqrt{3} \text{ km}$

(iii) $\angle N_2CA = \angle BAC = 30^\circ$ (Alternate angles on parallel north lines)

$\angle DCA = 30^\circ + 45^\circ$ (D is NW of C)
 $= 75^\circ$

$DA^2 = 8^2 + (10\sqrt{3})^2 - 2(8)(10\sqrt{3}) \cos 75^\circ$
 $= 292.2739$

$DA = 17.0960$

$\approx 17.1 \text{ km}$ (nearest 100 m)

$$b) (i) \quad y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1) \quad A(2, 8) \quad B(-6, -4)$$

$$y - 8 = \frac{-4 - 8}{-6 - 2} (x - 2)$$

$$y - 8 = \frac{3}{2} (x - 2)$$

$$2y - 16 = 3x - 6$$

$$3x - 2y + 10 = 0$$

$$(ii) \quad D: x = 0$$

$$3(0) - 2y + 10 = 0$$

$$2y = 10$$

$$y = 5$$

$$\therefore D(0, 5)$$

$$(iii) \quad C(5, 3), \quad 3x - 2y + 10 = 0$$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|3 \times 5 + (-2) \times 3 + 10|}{\sqrt{3^2 + (-2)^2}}$$

$$= \frac{19}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$$

$$= \frac{19\sqrt{13}}{13} \text{ units}$$

$$(iv) \quad d = \sqrt{(2 - (-6))^2 + (8 - (-4))^2}$$

$$= \sqrt{8^2 + 12^2}$$

$$= \sqrt{208}$$

$$= 4\sqrt{13} \text{ units.}$$

$$(v) \quad A = \frac{1}{2} \times 4\sqrt{13} \times \frac{19\sqrt{13}}{13}$$

$$= \frac{2 \times 19 \times 13}{13}$$

$$= 38 \text{ u}^2$$

$$c) (i) \quad L = r\theta$$

$$2\pi = 6\theta$$

$$\theta = \frac{\pi}{3}$$

$$(ii) \quad A = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 6 \times 6 \times \sin \frac{\pi}{3}$$

$$= 18 \times \frac{\sqrt{3}}{2}$$

$$= 9\sqrt{3} \text{ u}^2$$

13.a (i) $f(x) = -x^3 + \frac{3x^2}{2} + 6x$

$$f'(x) = -3x^2 + 3x + 6$$

$$f''(x) = -6x + 3$$

Stationary points $f'(x) = 0$

$$0 = -3x^2 + 3x + 6$$

$$0 = x^2 - x - 2$$

$$0 = (x-2)(x+1)$$

$$\therefore x = -1, 2$$

At $x = -1$, $y = -(-1)^3 + \frac{3(-1)^2}{2} + 6(-1)$

$$= -3.5$$

$$f''(x) = -6(-1) + 3$$

$$> 0$$

$\therefore (-1, -3.5)$ is a local minimum

At $x = 2$, $y = -(2)^3 + \frac{3(2)^2}{2} + 6(2)$

$$= 10$$

$$f''(x) = -6(2) + 3$$

$$< 0$$

$\therefore (2, 10)$ is a local maximum

(ii) Point of inflexion $f''(x) = 0$

$$0 = -6(x) + 3$$

$$-3 = -6x$$

$$x = \frac{1}{2}$$

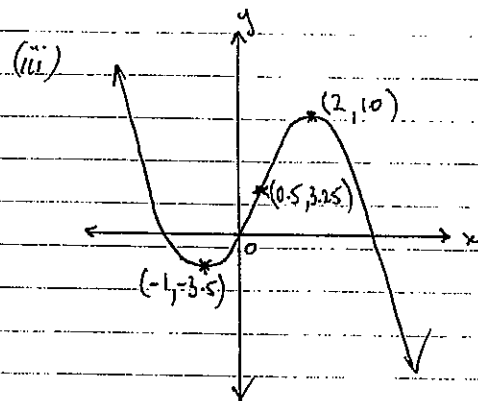
$$y = -\left(\frac{1}{2}\right)^3 + \frac{3\left(\frac{1}{2}\right)^2}{2} + 6\left(\frac{1}{2}\right)$$

$$= 3.25$$

$$(0.5, 3.25)$$

Check:

x	$\frac{1}{2}^-$	$\frac{1}{2}$	$\frac{1}{2}^+$
$f''(x)$	$+$	0	$-$



(iv) $x < -1$ and $x > 2$

b) $f(x) = 4 \sin 2\pi x$

(i) Amplitude = 4

Period = 1

(ii) $f(x) = 4 \sin 2\pi x$

$$f'(x) = 8\pi \cos 2\pi x$$

$$f''(x) = -16\pi^2 \sin 2\pi x$$

$$f'(x) = 0$$

$$0 = 8\pi \cos 2\pi x$$

$$0 = \cos 2\pi x$$

$$\cos \theta = 0$$

$$\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}$$

$$2\pi x = \frac{\pi}{2}$$

$$x = \frac{1}{4}$$

$$f''(x) = -16\pi^2 \sin\left(2\pi\left(\frac{1}{4}\right)\right)$$

$$< 0$$

$\therefore$ maximum

$$2\pi x = \frac{3\pi}{2}$$

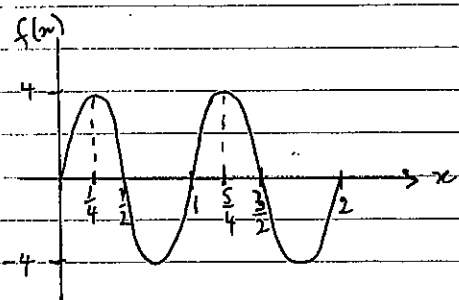
$$x = \frac{3}{4}$$

$$f''(x) = -16\pi^2 \sin\left(2\pi\left(\frac{3}{4}\right)\right)$$

Period = 1

Local maximums at $x = \frac{1}{4}, 1 + \frac{1}{4}$

(ii)



$$x = \frac{1}{4}, \frac{5}{4}$$

$$c) \lim_{x \rightarrow 2} \frac{3x^2 - 12}{x - 2} = \lim_{x \rightarrow 2} \frac{3(x+2)(x-2)}{x-2}$$

$$= 3(2+2)$$

$$= 12$$

$$14 a) \int_1^5 \frac{1}{x^2 + 1} dx$$

x	1	2	3	4	5
f(x)	0.5	0.2	0.1	$\frac{1}{17}$	$\frac{1}{26}$

$$\int_1^5 \frac{1}{x^2 + 1} dx \approx \frac{1}{3} \left[0.5 + 4 \left(0.2 + \frac{1}{17} \right) + 2 \left(0.1 \right) + \frac{1}{26} \right]$$

$$= 0.591251 \dots$$

$$= 0.5913 \text{ (4 d.p.)}$$

b) (i) 2nd card.

	3	4	5	6
3	X	34	35	36
4	43	X	45	46
5	53	54	X	56
6	63	64	65	X

$$(ii) P(W) = \frac{1}{4}$$

(iii) Start with \$5.

5 turns cost $5 \times \$1$

\$0 remaining

1 win $\approx +\$3$: 4 losses

$$P(1 \text{ win, 4 losses}) = \frac{1}{4} \times \left(\frac{3}{4} \right)^4 \times 5$$

$$= \frac{405}{1024}$$

$$c) A = \int_0^2 e^{\frac{1}{2}x} dx - \int_0^2 x dx$$

$$= \left[2e^{\frac{1}{2}x} \right]_0^2 - \left[\frac{x^2}{2} \right]_0^2$$

$$= \left(2e^{\frac{1}{2}(2)} - 2e^{\frac{1}{2}(0)} \right) - \left(\frac{(2)^2}{2} - \frac{(0)^2}{2} \right)$$

$$= 2e - 2 - 2 + 0$$

$$= 2e - 4$$

$$= 2(e-2) \text{ u}^2$$

$$d) 16y = x^2 - 4x - 12$$

$$(i) 16y = (x-2)^2 - 4 - 12$$

$$16y + 16 = (x-2)^2$$

$$16(y+1) = (x-2)^2$$

Vertex at $(2, -1)$

$$(ii) 4a = 16$$

$$a = 4$$

$$\therefore \text{Focus} = (2, -1+4) \quad (\text{Concave up})$$

$$= (2, 3)$$

$$(iii) \text{Directrix } y = -1-4$$

$$y = -5$$

$$15. a) (i) N = Ae^{kt}$$

$$t=0 \quad N = 7.13 \times 10^6 = A$$

$$t=3 \quad N = 7.3 \times 10^6$$

$$At \quad t=3$$

$$7.3 \times 10^6 = 7.13 \times 10^6 e^{3k}$$

$$e^{3k} = \frac{7.3}{7.13}$$

$$3k = \ln\left(\frac{7.3}{7.13}\right)$$

$$k = \ln\left(\frac{7.3}{7.13}\right)$$

$$3$$

$$= 0.0078543...$$

$$\approx 0.79\% \quad (2 \text{ d.p.})$$

$$(ii) 2019 \Rightarrow t=10$$

$$N = 7.13 \times 10^6 e^{0.0078 \times 10}$$

$$= 7.7 \times 10^6 \quad (\text{Nearest hundred thousand})$$

$$(iii) 10 \times 10^6 = 7.13 \times 10^6 e^{0.0079t}$$

$$e^{0.0079t} = \frac{10}{7.13}$$

$$t = \ln\left(\frac{10}{7.13}\right)$$

$$0.0079$$

$$= 43.068...$$

$\therefore$ During the 43 (2009 + 43) year

$\therefore$ During 2052.

b) (i) C

(ii) At 4s, it is in a positive direction from the origin (right) and has a negative velocity (moving towards the origin).

(iii) Between points A and D.

The graph is concave down during this time.

c) $4 \cos^2 \theta = 6 \sin \theta + 6$

$$4(1 - \sin^2 \theta) = 6 \sin \theta + 6$$

$$4 - 4 \sin^2 \theta - 6 \sin \theta - 6 = 0$$

$$-4 \sin^2 \theta - 6 \sin \theta - 2 = 0$$

Let $x = \sin \theta$

$$-4x^2 - 6x - 2 = 0$$

$$2x^2 + 3x + 1 = 0$$

$$(2x+1)(x+1) = 0$$

$$\therefore x = -\frac{1}{2}, -1$$

$$\sin \theta = -\frac{1}{2} \quad \left(\sin \frac{\pi}{6} = \frac{1}{2} \right)$$

$$\theta = \underline{\pi + \frac{\pi}{6}}, \underline{2\pi - \frac{\pi}{6}}, \underline{3\pi + \frac{\pi}{6}}, \underline{4\pi - \frac{\pi}{6}}, \dots$$

$$\sin \theta = -1 \quad \left(\sin \frac{\pi}{2} = 1 \right)$$

$$\theta = \underline{\pi + \frac{\pi}{2}}, \underline{3\pi + \frac{\pi}{2}}, \dots$$

But $0 \leq \theta < 2\pi$

$$\therefore \theta = \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$d) \frac{1}{\sqrt{n} + \sqrt{n+1}} \times \frac{\sqrt{n} - \sqrt{n+1}}{\sqrt{n} - \sqrt{n+1}} = \frac{\sqrt{n} - \sqrt{n+1}}{n - (n+1)}$$

$$= \frac{\sqrt{n} - \sqrt{n+1}}{-1}$$

$$= \sqrt{n+1} - \sqrt{n}$$

16. a) (i) $(\sqrt{3})^2 = x^2 + y^2$

$$y = \sqrt{(\sqrt{3})^2 - x^2}$$

$$= \sqrt{3 - x^2}$$

(ii) $L = 2x + 2y + \sqrt{3}$

$$= 2(x + y + \frac{\sqrt{3}}{2})$$

Sub $y = \sqrt{3 - x^2}$

$$L = 2(x + \sqrt{3 - x^2} + \frac{\sqrt{3}}{2})$$

(iii)

$$L = 2(x + (3 - x^2)^{\frac{1}{2}} + \frac{\sqrt{3}}{2})$$

$$\frac{dL}{dx} = 2(1 + \frac{1}{2}(3 - x^2)^{-\frac{1}{2}} \cdot 2x)$$

$$= 2 - \frac{2x}{\sqrt{3 - x^2}}$$

Stationary point: $\frac{dL}{dx} = 0$

$$0 = 2 + \frac{2x}{\sqrt{3 - x^2}}$$

$$\frac{2x}{\sqrt{3 - x^2}} = -2$$

$$2x = -2\sqrt{3 - x^2}$$

$$4x^2 = 4(3 - x^2)$$

$$4x^2 = 12 - 4x^2$$

$$8x^2 = 12$$

$$x^2 = \frac{3}{2}$$

$$x = \pm \sqrt{\frac{3}{2}}$$

But $x > 0$ (length)

$$\therefore x = \sqrt{\frac{3}{2}} \text{ m}$$

Check it's a maximum $\frac{d^2L}{dx^2} < 0$

$$\frac{dL}{dx} = 2 - 2x(3 - x^2)^{-\frac{1}{2}}$$

$$\frac{d^2L}{dx^2} = -2x \cdot (-\frac{1}{2})(-2x)(3 - x^2)^{-\frac{3}{2}} + (-2)(3 - x^2)^{-\frac{1}{2}}$$

$$= -2x^2(3 - x^2)^{-\frac{3}{2}} - 2(3 - x^2)^{-\frac{1}{2}}$$

$$= -\frac{2x^2}{\sqrt{(3 - x^2)^3}} - \frac{2}{\sqrt{(3 - x^2)}}$$

When $x = \sqrt{\frac{3}{2}}$

$$\frac{d^2L}{dx^2} = -\frac{2(\frac{3}{2})}{\sqrt{(3 - \frac{3}{2})^3}} - \frac{2}{\sqrt{(3 - \frac{3}{2})}}$$

$$= -3.265$$

$$< 0$$

$\therefore$ maximum

$\therefore$ Maximum strength when $x = \sqrt{\frac{3}{2}}$

b) (i) In $\triangle AEB$,

$$\angle AEB = \angle ABE \quad (\text{Equal angles opposite equal sides})$$

$$AE = AB \quad (\text{Given})$$

$$\therefore 2\angle ABE + 30 = 180 \quad (\angle \text{sum of } \triangle \text{ is } 180)$$

$$\therefore \angle ABE = 75^\circ$$

But

$$\angle ABE + \angle ABC = 138^\circ \quad (\angle EBC = 138^\circ \text{ (Given)})$$

$$\therefore \angle ABC = 138 - 75$$

$$= 63^\circ$$

$$\angle ABC + \angle BCD = 63 + 83 + 34$$

$$= 180$$

$$\therefore AB \parallel DC \quad (\text{Co-interior angles add to } 180^\circ)$$

(ii) In $\triangle ABC$ and $\triangle CDA$,

$$A \quad \angle BAC = \angle DCA = 34^\circ \quad (\text{Alternate angles on } AB \parallel DC)$$

$$A \quad \angle BCA = \angle DAC = 83^\circ \quad (\text{Alternate angles on } AD \parallel BC)$$

$$S \quad AC \text{ is common}$$

$$\therefore \triangle ABC \equiv \triangle CDA \quad (\text{A.A.S.})$$

$$c) \quad V = \pi \int_0^{\frac{\pi}{4}} \sec^2 x \, dx$$

$$= \pi \left[\tan x \right]_0^{\frac{\pi}{4}}$$

$$= \pi \left[\tan \frac{\pi}{4} - \tan 0 \right]$$

$$= \pi \cdot 1^3$$